

Ubiquinone Biosynthesis from Chorismate-Derived 4-Hydroxybenzoate: A Mechanistic Review

A review-style synthesis of the de novo pathway to coenzyme Q, from the aromatic head group 4-hydroxybenzoate and a polyprenyl side chain through the fully substituted benzoquinone ring.

1. Executive summary

Ubiquinone (coenzyme Q, UQ or CoQ) is a redox-active polyprenylated benzoquinone lipid that shuttles electrons and protons in respiratory and photosynthetic electron-transport chains and acts as a membrane antioxidant. Its biosynthesis is conceptually simple — take an aromatic ring, hang a long isoprenoid tail on it, and decorate the ring with two methoxy groups, one methyl group, and two ring-oxygens arranged as a *para*-quinone — but mechanistically remarkable, because almost every tailoring reaction operates on an extremely hydrophobic substrate.

The best-understood version of the pathway, in *Escherichia coli*, proceeds in five conceptual modules: (1) **head-group synthesis**, in which UbiC (chorismate lyase) converts the shikimate-pathway branch-point metabolite chorismate to 4-hydroxybenzoate (4-HB); (2) **prenylation**, in which the intramembrane prenyltransferase UbiA attaches an octaprenyl tail to 4-HB; (3) **decarboxylation**, in which the UbiD family enzyme removes the ring carboxyl using the exotic prenylated-FMN (prFMN) cofactor supplied by the flavin prenyltransferase UbiX; and (4–5) **ring tailoring**, three hydroxylations (UbiI/UbiH/UbiF, FAD-monooxygenases) interleaved with three methylations (the O-methyltransferase UbiG and the C-methyltransferase UbiE). A central modern insight is that in bacteria the tailoring reactions are carried out not by free membrane enzymes but by a soluble ~1-MDa **Ubi metabolon** that sequesters the hydrophobic intermediates in the SCP2 lipid-binding cavity of UbiJ [PMID:30686758]. Eukaryotes reproduce this metabolon logic on the matrix face of the mitochondrial inner membrane with a **COQ**

"**synthome**" (Coq3–Coq9 plus the regulatory UbiB-family kinase Coq8) [PMID:17391640; [P 38425362](#)],

but with a divergent parts list — notably lacking recognizable UbiC/UbiX/UbiD orthologs.

The system's deepest origin appears to lie in **cyanobacteria**, with subsequent diversification in proteobacteria and, via the mitochondrial endosymbiont, in eukaryotes [PMID:29106540]. The clearest axis of variation is **oxygen**: the canonical hydroxylases are O₂-dependent flavin monooxygenases, whereas many bacteria run an **O₂-independent route** (UbiT/UbiU/UbiV) to make UQ anaerobically [PMID:31289180; [P 32409583](#)].

Below I lay out the boundaries of the system, the current mechanistic model, the molecular machines that execute it, its evolutionary and cell-biological variants, the ordering and compartment constraints, and the principal controversies.

2. Definition and biological boundaries

2.1 What is included

For the purpose of this review, the "system" is the **de novo enzymatic conversion of 4-hydroxybenzoate plus a polyprenyl-diphosphate into mature ubiquinone**, comprising:

- Formation of the aromatic head group 4-HB from chorismate (UbiC in γ -proteobacteria).
- Attachment of the polyprenyl tail (UbiA / eukaryotic COQ2).
- Removal of the ring carboxyl (UbiD + UbiX cofactor supply).
- Sequential ring hydroxylation and O-/C-methylation to install the two methoxy, one methyl and the *para*-quinone oxygens (UbiI/H/F, UbiG, UbiE; eukaryotic Coq6/Coq7, Coq3, Coq5).
- The accessory factors that organize these reactions (UbiJ/UbiK scaffolds; Coq4/Coq8/Coq9).

2.2 Adjacent processes that should be treated separately

Several neighboring processes are frequently conflated with UQ biosynthesis and are better treated as **inputs, outputs, or parallel pathways**:

- **The shikimate pathway and chorismate branch point.** Chorismate is the shared precursor of folate, menaquinone, enterobactin, tryptophan, tyrosine and phenylalanine.

UbiC merely taps this pool; the shikimate pathway itself is upstream and not part of UQ biosynthesis proper.

- **Isoprenoid (polyprenyl-diphosphate) synthesis.** The tail is built by the MEP or mevalonate pathway plus a dedicated polyprenyl-diphosphate synthase (IspB in *E. coli*; PDSS1/PDSS2 in humans). This determines side-chain length (Q_8 in *E. coli*, Q_6 in yeast, Q_9/Q_{10} in mammals) but is a **separate module** feeding UbiA.
- **Menaquinone / plastoquinone biosynthesis.** These are chemically related prenylquinones made by partly homologous machinery (the UbiA superfamily of intramembrane prenyltransferases is shared) [PMID:26922674;

P 28018418

but they use different ring precursors and are distinct pathways.

- **The mevalonate pathway and its clinical modulation.** Statins lower CoQ indirectly by blocking mevalonate-derived isoprenoid supply; this is a regulatory/pharmacological effect on precursor availability, not a step of the ring-modification pathway [PMID:37202505].
- **Downstream UQ function and trafficking.** UQ's roles in respiration, in ferroptosis suppression via FSP1 [PMID:35395704], and its inter-organelle distribution (mediated in yeast by the UbiB-family proteins Cqd1/Cqd2) [PMID:34362905] are consequences of, not steps in, biosynthesis.

2.3 Competing definitions

There is genuine definitional tension over **where the ring precursor comes from** and therefore where the pathway "starts." The chorismate→4-HB→UbiC definition is faithful to *E. coli*, but is not universal: eukaryotes generate 4-HB largely from **tyrosine** (and some fungi via a **mandelate pathway**) rather than by a UbiC-type chorismate lyase [PMID:32561586;

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and several organisms can also use alternative ring precursors (e.g., *para*-aminobenzoate in yeast, coumarate/vanillate in some plants). A species-neutral definition therefore treats "4-HB (or an equivalent aromatic ring precursor) entering prenylation" as the true boundary, and treats "chorismate → 4-HB by UbiC" as the *bacterial-specific* entry reaction.

3. Mechanistic overview

The consensus reaction sequence, using *E. coli* (aerobic) nomenclature, is:

1. **Chorismate → 4-hydroxybenzoate (4-HB)** — UbiC chorismate (pyruvate-)lyase eliminates pyruvate and aromatizes the ring [PMID:1644192;

P 4595202

Obligatory as an entry point in bacteria; substituted by tyrosine-derived routes in eukaryotes.

2. **4-HB → 3-octaprenyl-4-hydroxybenzoate (OPP)** — UbiA transfers the octaprenyl tail from octaprenyl-diphosphate onto ring C3 [PMID:4552989;

P 26922674

Obligatory; commits the ring to the membrane phase.

3. **OPP → 2-octaprenylphenol (2-OPP)** — UbiD decarboxylates the ring using prFMN; UbiX supplies prFMN by prenylating FMN with a dimethylallyl unit [PMID:28754323;

P 32951834

Obligatory in bacteria; UbiX cofactor supply is obligately upstream of UbiD.

4. **Ring tailoring — three hydroxylations + three methylations, interleaved.** The ring is decorated by three FAD-monooxygenase hydroxylations at distinct ring positions — UbiI catalyzes the C5-hydroxylation [PMID:23709220], with UbiH and UbiF acting at the remaining positions — alternating with O- and C-methylations, because each O-methylation requires a hydroxyl installed by a preceding monooxygenase step. The two O-methyl transfers are both catalyzed by the single SAM-dependent O-methyltransferase **UbiG** — a bifunctional assignment demonstrated directly by showing that purified *E. coli* UbiG (and its eukaryotic ortholog Coq3) catalyzes both O-methylation steps [PMID:10419476] — and the single ring C-methyl by **UbiE** [PMID:24480387]. (Exact positional order of individual tailoring steps is best defined in *E. coli*; the precise sequence and intermediate identities differ in eukaryotes.)

The **best current model** is thus a defined, largely linear reaction order in which prenylation and decarboxylation precede the alternating hydroxylation/methylation steps, and in which the whole tailoring phase is executed within a supramolecular assembly rather than by diffusing enzymes.

Obligatory vs conditional vs accessory: - *Obligatory chemistry:* prenylation (UbiA), decarboxylation (UbiD/UbiX), and the full set of three ring-oxygen insertions and three methylations — no mature UQ forms without them. - *Conditional:* the **identity of the hydroxylases** depends on oxygen. Under aerobiosis, FAD monooxygenases (UbiI/UbiH/UbiF) use O₂ as the oxygen donor; under anaerobiosis, UbiT/UbiU/UbiV replace them with an O₂-independent mechanism [PMID:23709220;

P 31289180

P 32409583

UbiC itself is conditional/replaceable (tyrosine route in eukaryotes). - *Accessory:* the scaffolding/regulatory factors **UbiJ**, **UbiK** (bacteria) [PMID:36142227;

P 28559279

and **Coq4**, **Coq8**, **Coq9** (eukaryotes) [PMID:17391640;

P 38425362

are essential for efficient flux and complex integrity but do not themselves carry out ring chemistry (Coq8 is an ATPase/atypical kinase, not a canonical Coq enzyme).

4. Major molecular players and active assemblies

Module	Bacterial enzyme (family)	Reaction	Eukaryotic counterpart	Key refs
Head group	UbiC (chorismate lyase)	chorismate → 4-HB + pyruvate	(no ortholog; 4-HB from Tyr)	P 1644192
Prenylation	UbiA (UbiA intramembrane prenyltransferase superfamily)	4-HB + octaprenyl-PP → OPP	CoQ2	P 4552989 ; P 26922674
Cofactor supply	UbiX (flavin prenyltransferase)	FMN + DMAP(P) → prFMN	(none)	P 31142738 ; P 32951834
Decarboxylation	UbiD (prFMN (de)carboxylase)	OPP → 2-OPP + CO ₂	(non-orthologous)	P 28754323 ; P 33763291
Hydroxylation	UbiI , UbiH , UbiF (FAD monooxygenases, aerobic) / UbiU , UbiV (anaerobic)	3× ring hydroxylation	Coq6 (+Coq7 di-iron hydroxylase)	P 23709220 ; P 31289180
Methylation	UbiG (O-MTase), UbiE (C-MTase), SAM-dependent	2× O-methyl, 1× C-methyl	Coq3 (O), Coq5 (C)	P 24480387
Scaffold / regulation	UbiJ (SCP2 cavity), UbiK	substrate channeling	Coq4 , Coq8 /ADCK, Coq9	P 30686758 ; P 36142227 ; P 38425362

4.1 The prFMN cofactor and the UbiX–UbiD module

UbiX performs a non-metal-dependent prenyl transfer resembling class I terpene-cyclase chemistry, linking a dimethylallyl moiety to the FMN N5 and C6 to create a fourth, non-aromatic ring; after oxidative maturation to an iminium species, prFMN is competent for UbiD catalysis [PMID:31142738;

P 32951834

UbiD-family enzymes are **reversible (de)carboxylases**; the well-characterized model member ferulic acid decarboxylase (Fdc1) reveals a reversible **1,3-dipolar cycloaddition** between prFMN and substrate [PMID:33763291;

P 28754323

This module is the mechanistic showpiece of the pathway and the part with the greatest biocatalytic reach beyond UQ (aromatic acid (de)carboxylation) [PMID:38176649].

4.2 The Ubi metabolon and the hydrophobicity problem

The signature organizational insight is that the last six reactions in *E. coli* are catalyzed by a **stable, soluble ~1-MDa metabolon of seven Ubi proteins**, in which the SCP2 domain of **UbiJ** provides an extended hydrophobic cavity that binds the polyprenylated intermediates and hands them from enzyme to enzyme in a hydrophilic cytoplasm [PMID:30686758]. **UbiK** is an accessory factor that complexes with UbiJ [PMID:28559279;

P 36142227

This directly resolves the paradox of how soluble tailoring enzymes act on membrane-partitioning substrates.

4.3 The eukaryotic COQ synthome

Yeast Coq3/4/5/6/7/9 co-migrate at ~1 MDa and are mutually stabilizing (loss of one destabilizes the others) [PMID:17391640]. Reconstitution of the animal metabolon in vitro (via ancestral sequence reconstruction) captured the pathway and demonstrated that the atypical **UbiB-family kinase COQ8** "increases and streamlines" CoQ output [PMID:38425362]. COQ8/ADCK3/ADCK4 is an ancient atypical protein-kinase-like ATPase whose activity is stimulated by CoQ intermediates and phospholipids and is required for synome integrity [PMID:32479562;

P 36302899

5. Evolutionary and cell-biological variation

5.1 Across lineages

- **Deep origin.** Comparative genomics places the origin of Q biosynthesis in **cyanobacteria**, followed by diversification in anaerobic α -proteobacteria whose extant relatives resemble the mitochondrial pathway more than the *E. coli* Ubi pathway [PMID:29106540]. Two branches emerge: the *ubi* genes (proteobacteria/*E. coli*) and the *coq* genes (eukaryotes/mitochondria).
- **Head-group entry differs.** Chorismate lyase (UbiC) is essentially a bacterial solution; eukaryotes derive 4-HB from **tyrosine**, and some ascomycete yeasts use a **mandelate pathway** for benzenoid supply [PMID:32561586;

P 30822625

- **The prFMN module is bacterially specific.** Eukaryotes lack recognizable UbiX/UbiD, so the decarboxylation and its cofactor supply are handled by **non-orthologous** means — a major discontinuity in an otherwise deeply conserved pathway.
- **Side-chain length is lineage-tuned** by the polyprenyl-diphosphate synthase (Q₈ *E. coli*, Q₆ *S. cerevisiae*, Q₉ rodents, Q₁₀ humans), even though the ring chemistry is conserved.

5.2 Physiological-state variation (oxygen)

The strongest, best-supported conditional variation is the **oxygen regime**. Aerobically, three FAD monooxygenases (UbiI/UbiH/UbiF) insert ring oxygens using O₂ [PMID:23709220]. Anaerobically, a distinct set (**UbiT, UbiU, UbiV**) performs O₂-independent hydroxylation, allowing continuous UQ production across the whole O₂ range [PMID:31289180;

P 32409583

UbiT carries an SCP2 domain (a UbiJ paralog) specialized for anaerobic conditions. This is a textbook case of **alternative molecular routes achieving the same chemical outcome**.

5.3 Compartment and cell biology

- In bacteria, evidence points to a **cytoplasmic/soluble metabolon** despite lipid substrates [PMID:30686758].
- In eukaryotes, synthesis occurs on the **matrix face of the mitochondrial inner membrane** via the COQ synthome [PMID:17391640], after which UQ must be **distributed** to other

membranes — a process governed by UbiB-family proteins (Cqd1/Cqd2 in yeast) and still incompletely understood [PMID:34362905].

- **Disease relevance:** defects in COQ genes cause primary CoQ deficiency (e.g., steroid-resistant nephrotic syndrome, encephalomyopathy), sometimes responsive to CoQ₁₀ supplementation [PMID:35643375;

P 39864756

underscoring that the pathway's output is rate-limiting for human mitochondrial function.

6. Constraints, dependencies, and failure modes

- **Strict ordering at the front end.** Prenylation (UbiA) must precede decarboxylation and tailoring: the ring must be committed to the lipid phase before the hydrophobic-substrate-handling machinery engages it. UbiC (or an equivalent 4-HB source) must precede UbiA.
- **Cofactor-supply dependency.** UbiX-generated prFMN is *obligately* required before UbiD can act; without mature prFMN, decarboxylation fails and prenylphenol intermediates accumulate [PMID:28754323].
- **Alternating hydroxylation/methylation.** The interleaving of O-methylation (which requires a preceding hydroxyl) and hydroxylation imposes a defined sequence; the two UbiG O-methylations act on hydroxyls generated by preceding monooxygenase steps [PMID:24480387].
- **Oxygen as a hard constraint on enzyme identity.** Aerobic FAD monooxygenases cannot function without O₂; organisms that respire anaerobically *must* deploy the UbiT/U/V system or fail to make UQ under anoxia [PMID:31289180]. UbiI is demonstrably aerobic-only, with an alternative enzyme catalyzing C5-hydroxylation anaerobically [PMID:23709220].
- **Metabolon integrity as a failure mode.** In both bacteria and eukaryotes, disrupting a scaffold or one enzyme destabilizes the whole assembly, producing a CoQ-less phenotype and accumulation of pathway intermediates [PMID:17391640;

P 36142227

This is the molecular basis of secondary CoQ deficiencies.

- **Evidence ruling out otherwise-plausible paths.** The metabolon data argue *against* a purely diffusion-based, fully membrane-embedded assembly line for the tailoring steps

[PMID:30686758]; the identification of the O₂-independent hydroxylases rules against the earlier assumption that UQ synthesis is strictly aerobic [PMID:31289180].

7. Controversies and open questions

1. **Do bacteria run the pathway soluble or membrane-bound?** The ~1-MDa soluble Ubi metabolon challenges the classical membrane-bound model [PMID:30686758]. How intermediates partition between the SCP2 cavity and the membrane, and whether the metabolon docks transiently to membranes, remain open.
2. **How is the anaerobic hydroxylation chemistry actually done?** UbiT/U/V enable O₂-independent hydroxylation, but the oxygen donor and catalytic mechanism (proposed metallo-/radical chemistry) are not fully resolved [PMID:32409583].
3. **What replaces UbiC/UbiX/UbiD in eukaryotes?** The eukaryotic decarboxylation step and its "cofactor logic" are enigmatic; several reactions in the animal metabolon were only recently captured in vitro and some remain uncharacterized [PMID:38425362]. More broadly, as of authoritative review the CoQ pathway "has not been fully defined in any organism," with multiple steps catalyzed by still-unidentified enzymes [PMID:28927698].
4. **What does COQ8 actually do?** Despite being essential, the precise catalytic contribution of the UbiB-family kinase/ATPase (phosphorylation vs ATP-dependent chaperoning of the synome, vs lipid extraction) is still debated [PMID:32479562;

P 36302899

5. **How is mature UQ trafficked from its synthesis site?** The export/distribution machinery (Cqd1/Cqd2 and beyond) is only beginning to be defined [PMID:34362905].
6. **Cross-organism extrapolation risks.** Much mechanistic detail comes from *E. coli*, *S. cerevisiae*, and reconstituted animal enzymes; ring-modification order, precursor choice, and enzyme complements differ enough between lineages that findings should not be assumed universal.

8. Key references

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Uncertainty statement. The bacterial pathway (especially *E. coli*) is the most rigorously supported, including enzyme structures and reconstituted chemistry for UbiX/UbiD and the metabolon. Eukaryotic steps are strong for enzyme identity and complex organization but weaker for the decarboxylation-equivalent chemistry and for UQ trafficking. Oxygen-dependent vs O₂-independent hydroxylation is well supported genetically/biochemically, but the anaerobic mechanism is not fully solved. Claims here are anchored to specific organisms and should not be assumed to hold uniformly across all of biology.
